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### Cold brew devices, systems and methods

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#### Abstract

Versatile devices, systems, and methods of cold brewing coffee and/or tea using pre-manufactured disposable coffee-containing cups. Alternatively, coffee or tea may be placed within a meshed container for cold brew. A single mug device and large volume dispensers using the same cold brew technique, which has been shown to produce higher dissolved coffee/tea, resulting in a higher quality beverage.

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name   | U.S. Cl. | CPC         |
|--------------|-------------|-----------------|----------|-------------|
| D661544      | 12/2011     | Enghard         | N/A      | N/A         |
| 8307755      | 12/2011     | Shen            | N/A      | N/A         |
| 8985377      | 12/2014     | Lane            | N/A      | N/A         |
| 9089239      | 12/2014     | Billadeau       | N/A      | N/A         |
| D747140      | 12/2015     | Roth et al.     | N/A      | N/A         |
| 9314126      | 12/2015     | Molayem         | N/A      | N/A         |
| 9332873      | 12/2015     | Tien            | N/A      | N/A         |
| 9510705      | 12/2015     | Rolfes et al.   | N/A      | N/A         |
| 9801492      | 12/2016     | Lin             | N/A      | N/A         |
| 9877607      | 12/2017     | Beckman et al.  | N/A      | N/A         |
| 9901167      | 12/2017     | Anderson        | N/A      | N/A         |
| D815895      | 12/2017     | Roth et al.     | N/A      | N/A         |
| D823052      | 12/2017     | Neace, Jr.      | N/A      | N/A         |
| D823053      | 12/2017     | Neace, Jr.      | N/A      | N/A         |
| 10149568     | 12/2017     | Shen            | N/A      | N/A         |
| 10299633     | 12/2018     | Shaw            | N/A      | N/A         |
| 10405688     | 12/2018     | Lin             | N/A      | N/A         |
| 10463190     | 12/2018     | Hambly et al.   | N/A      | N/A         |
| D876887      | 12/2019     | Barber          | N/A      | N/A         |
| 10588448     | 12/2019     | Lin             | N/A      | N/A         |
| D882338      | 12/2019     | Bellman         | N/A      | N/A         |
| 2010/0263549 | 12/2009     | Lee             | N/A      | N/A         |
| 2014/0072684 | 12/2013     | Madden          | 426/431  | A47J 31/005 |
| 2014/0224725 | 12/2013     | Uspenski et al. | N/A      | N/A         |
| 2015/0175347 | 12/2014     | Empl            | 426/433  | A23C 9/1508 |
| 2015/0282664 | 12/2014     | Savage          | 426/431  | A23F 5/262  |
| 2017/0265674 | 12/2016     | Liu             | N/A      | N/A         |

|              |         |               |     |              |
|--------------|---------|---------------|-----|--------------|
| 2017/0334636 | 12/2016 | Park          | N/A | B65D 85/8043 |
| 2019/0002192 | 12/2018 | Wicks         | N/A | A47J 31/407  |
| 2019/0200800 | 12/2018 | Redington     | N/A | N/A          |
| 2020/0008605 | 12/2019 | Maloni et al. | N/A | N/A          |
| 2020/0039738 | 12/2019 | Duman         | N/A | A47G 19/2272 |
| 2020/0154930 | 12/2019 | Rivera        | N/A | A47J 31/4457 |
| 2020/0297153 | 12/2019 | Jacobs        | N/A | A47J 31/5253 |
| 2022/0125234 | 12/2021 | Richardson    | N/A | A23F 5/262   |

## FOREIGN PATENT DOCUMENTS

| Patent No.    | Application Date | Country | CPC |
|---------------|------------------|---------|-----|
| 105852646     | 12/2015          | CN      | N/A |
| 207768165     | 12/2017          | CN      | N/A |
| 109619945     | 12/2018          | CN      | N/A |
| 110960041     | 12/2019          | CN      | N/A |
| 202018103516  | 12/2018          | DE      | N/A |
| 3082716       | 12/2019          | FR      | N/A |
| 200364716     | 12/2003          | KR      | N/A |
| 20140082912   | 12/2013          | KR      | N/A |
| 200475003     | 12/2013          | KR      | N/A |
| WO 2019243397 | 12/2018          | WO      | N/A |

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## Background/Summary

(1) This Patent Application claims priority to U.S. Provisional Patent Application Ser. No. 63/199,890, filed Feb. 1, 2021; and to U.S. Provisional Patent Application Ser. No. 63/049,027, filed Jul. 7, 2020; and to U.S. Provisional Patent Application Ser. No. 63/021,681, filed May 8, 2020; the contents of which are hereby incorporated by reference herein in their entirety into this disclosure.

## BACKGROUND OF THE SUBJECT DISCLOSURE

### Field of the Subject Disclosure

(1) The present subject disclosure relates to cold brew devices, systems, and methods. More specifically, the present subject disclosure relates to devices, systems, and methods to produce high quality cold brewed coffee or tea for an individual size or in larger volumes.

### Background of the Subject Disclosure

(2) Coffee and tea consumption spans across every culture and corner of the globe. The traditional hot cup of coffee or tea is ubiquitous is a stable morning ritual in virtually all regions of the world. It provides a distinct and satisfying taste to billions of drinkers every day.

(3) Coffee or tea is typically brewed in a pot using hot water exposed to coffee grounds or tea leaves. In both cases, water or milk is contacted with the grounds or leaves, which are filtered from the brewed beverage so that the brewed beverage may be consumed without the grinds or leaves. More modern techniques have used single serving cups or containers that allow a brewing system to access the coffee or tea within the single serving cup, and then dispense the hot brewed beverage

into a serving container. These modern systems generally utilize disposable single serving containers which are pre-manufactured with a given volume of ground coffee or tea, which are suitable for the single use hot brew systems.

(4) A less conventional, but increasingly popular method for preparing coffee is the cold brew technique, which has become a popular alternative to the traditional hot coffee. Cold brew coffee offers a less acidic, more flavorful alternative to traditionally brewed hot coffee. In warm or hot weather, it is a perfectly refreshing beverage to consume in an outdoor environment.

(5) However, the conventional cold brew method of preparing coffee is very time consuming and requires costly special equipment, which does not provide the same convenience and flexibility of preparing a single cup of hot coffee.

#### SUMMARY OF THE SUBJECT DISCLOSURE

(6) The present subject disclosure introduces an easy to use, versatile, inexpensive, and portable option to create cold-brew coffee at home/office using widely available disposable single serve coffee cups. Although there are conventional systems available for making cold brew coffee at home, none provide the simple elegance and versatility of the present subject disclosure which, among other things, utilizes one or two disposable single serve coffee cups. The present subject disclosure provides specialized beverage containers, baskets, filters, and techniques which produce a superior quality beverage. The present subject disclosure uses the same unique techniques to describe countertop systems which produce cold brewed beverages at greater volumes than for a single service.

(7) In one exemplary embodiment, the present subject disclosure is a device. The device includes a container; a lid that is attachable to the container; and a basket securable to the lid and positioned inside the container, the basket adapted to hold coffee grinds or tea leaves.

(8) In another exemplary embodiment, the present subject disclosure is a system. The system includes a container; a basket securable to and positioned inside the container, the basket adapted to hold coffee grinds or tea leaves; and a dispenser adapted to remove brewed beverage from a bottom of the container.

(9) In another exemplary embodiment, the present subject disclosure is a method of brewing a beverage. The method includes providing a container with a certain volume of fluid, providing a basket containing raw material for a beverage; securing the basket inside the container such that all portions of the raw material are submerged within the fluid; and allowing the cyclic downward flow of fluid through the basket and through the raw material to create a brewed beverage at a bottom of the container.

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## Description

#### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A shows a side view of device with a lid for cold brew, according to an exemplary embodiment of the present subject disclosure.

(2) FIG. 1B shows a side view of a device without a lid for cold brew, according to an exemplary embodiment of the present subject disclosure.

(3) FIG. 1C shows a perspective view of device for cold brew, according to an exemplary embodiment of the present subject disclosure.

(4) FIG. 1D shows a top view of device for cold brew, according to an exemplary embodiment of the present subject disclosure.

(5) FIG. 1E shows a top view of lid, according to an exemplary embodiment of the present subject disclosure.

(6) FIG. 2A shows a perspective view of container, according to an exemplary embodiment of the present subject disclosure.

(7) FIG. 2B shows a side view of container, according to an exemplary embodiment of the present subject disclosure.

(8) FIG. 2C shows a top view of container, according to an exemplary embodiment of the present subject disclosure.

(9) FIG. 3A shows a shadow perspective view of a lid, according to an exemplary embodiment of the present subject disclosure.

(10) FIG. 3B shows a shadow top view of a lid, according to an exemplary embodiment of the present subject disclosure.

(11) FIG. 3C shows a bottom view of a lid, according to an exemplary embodiment of the present subject disclosure.

(12) FIG. 3D shows a top view of a lid, according to an exemplary embodiment of the present subject disclosure.

(13) FIG. 4A shows an upper perspective view of an upper side of a lid closing mechanism, according to an exemplary embodiment of the present subject disclosure.

(14) FIG. 4B shows a lower perspective view of an upper side of a lid closing mechanism, according to an exemplary embodiment of the present subject disclosure.

(15) FIG. 4C shows a bottom view of an upper side of a lid closing mechanism, according to an exemplary embodiment of the present subject disclosure.

(16) FIG. 4D shows an upper perspective view of a lower side of a lid closing mechanism, according to an exemplary embodiment of the present subject disclosure.

(17) FIG. 4E shows a top view of a lower side of a lid closing mechanism, according to an exemplary embodiment of the present subject disclosure.

(18) FIG. 4F shows a side view of a lower side of a lid closing mechanism, according to an exemplary embodiment of the present subject disclosure.

(19) FIG. 5A shows a perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(20) FIG. 5B shows a side view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(21) FIG. 5C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(22) FIG. 6A shows an exploded perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(23) FIG. 6B shows an exploded view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(24) FIG. 6C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(25) FIG. 7A shows a perspective view of a spacer and upper slicer, according to an exemplary embodiment of the present subject disclosure.

(26) FIG. 7B shows a side view of a spacer and upper slicer, according to an exemplary embodiment of the present subject disclosure.

(27) FIG. 7C shows a top view of a spacer and upper slicer, according to an exemplary embodiment of the present subject disclosure.

(28) FIG. 8A shows a perspective view of an upper holder and lower slicer, according to an exemplary embodiment of the present subject disclosure.

(29) FIG. 8B shows a side view of an upper holder and lower slicer, according to an exemplary embodiment of the present subject disclosure.

(30) FIG. 8C shows a top view of an upper holder and lower slicer, according to an exemplary embodiment of the present subject disclosure.

(31) FIG. 9A shows a perspective view of an upper or lower slicer, according to an exemplary embodiment of the present subject disclosure.

(32) FIG. 9B shows a side view of an upper or lower slicer, according to an exemplary embodiment of the present subject disclosure.

(33) FIG. 9C shows a top view of an upper or lower slicer, according to an exemplary embodiment of the present subject disclosure.

(34) FIG. 10A shows a perspective view of a lower holder, according to an exemplary embodiment of the present subject disclosure.

(35) FIG. 10B shows a side view of a lower holder, according to an exemplary embodiment of the present subject disclosure.

(36) FIG. 10C shows a top view of a lower holder, according to an exemplary embodiment of the present subject disclosure.

(37) FIG. 11A shows a perspective view of a cutting tool, according to an exemplary embodiment of the present subject disclosure.

(38) FIG. 11B shows a side view of a cutting tool, according to an exemplary embodiment of the present subject disclosure.

(39) FIG. 11C shows a top view of a cutting tool, according to an exemplary embodiment of the present subject disclosure.

(40) FIG. 12A shows an exploded perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(41) FIG. 12B shows an exploded view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(42) FIG. 12C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(43) FIG. 13A shows a perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(44) FIG. 13B shows an exploded view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(45) FIG. 13C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(46) FIG. 14A shows a perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(47) FIG. 14B shows an exploded view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(48) FIG. 14C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(49) FIG. 15A shows a perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(50) FIG. 15B shows an exploded view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(51) FIG. 15C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(52) FIG. 16A shows a perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(53) FIG. 16B shows an exploded view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(54) FIG. 16C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(55) FIG. 17A shows a perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(56) FIG. 17B shows an exploded view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(57) FIG. 17C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(58) FIG. 18A shows a perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(59) FIG. 18B shows an exploded view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(60) FIG. 18C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(61) FIG. 19A shows a perspective top view of a lid, according to an exemplary embodiment of the present subject disclosure.

(62) FIG. 19B shows a perspective bottom view of a lid, according to an exemplary embodiment of the present subject disclosure.

(63) FIG. 19C shows a bottom view of a lid, according to an exemplary embodiment of the present subject disclosure.

(64) FIG. 20A shows a perspective top view of a holder/slicer, according to an exemplary embodiment of the present subject disclosure.

(65) FIG. 20B shows a perspective bottom view of a holder/slicer, according to an exemplary embodiment of the present subject disclosure.

(66) FIG. 20C shows a bottom view of a holder/slicer, according to an exemplary embodiment of the present subject disclosure.

(67) FIG. 21A shows a perspective top view of a holder/slicer, according to an exemplary embodiment of the present subject disclosure.

(68) FIG. 21B shows a perspective bottom view of a holder/slicer, according to an exemplary embodiment of the present subject disclosure.

(69) FIG. 21C shows a bottom view of a holder/slicer, according to an exemplary embodiment of the present subject disclosure.

(70) FIG. 22A shows a perspective top view of a holder/slicer, according to an exemplary embodiment of the present subject disclosure.

(71) FIG. 22B shows a perspective bottom view of a holder/slicer, according to an exemplary embodiment of the present subject disclosure.

(72) FIG. 22C shows a bottom view of a holder/slicer, according to an exemplary embodiment of the present subject disclosure.

(73) FIG. 23A shows a perspective top view of a spacer/slicer, according to an exemplary embodiment of the present subject disclosure.

(74) FIG. 23B shows a perspective bottom view of a spacer/slicer, according to an exemplary embodiment of the present subject disclosure.

(75) FIG. 23C shows a bottom view of a spacer/slicer, according to an exemplary embodiment of the present subject disclosure.

(76) FIG. 24A shows a perspective view of an open basket device, according to an exemplary embodiment of the present subject disclosure.

(77) FIG. 24B shows a shadow side view of an open basket device, according to an exemplary embodiment of the present subject disclosure.

(78) FIG. 24C shows a top view of an open basket device, according to an exemplary embodiment of the present subject disclosure.

(79) FIG. 25A shows a perspective view of an open basket, according to an exemplary embodiment of the present subject disclosure.

(80) FIG. 25B shows a side view of an open basket, according to an exemplary embodiment of the present subject disclosure.

(81) FIG. 25C shows a top view of an open basket, according to an exemplary embodiment of the present subject disclosure.

(82) FIG. 25D shows a perspective view of an open basket holder, according to an exemplary embodiment of the present subject disclosure.

(83) FIG. 26A shows a perspective view of a basket lid, according to an exemplary embodiment of the present subject disclosure.

(84) FIG. 26B shows a side view of a basket lid, according to an exemplary embodiment of the present subject disclosure.

(85) FIG. 26C shows a top view of a basket lid, according to an exemplary embodiment of the present subject disclosure.

(86) FIG. 27A shows a perspective view of a closed basket, according to an exemplary embodiment of the present subject disclosure.

(87) FIG. 27B shows an exploded side view of a closed basket, according to an exemplary embodiment of the present subject disclosure.

(88) FIG. 27C shows a top view of a closed basket, according to an exemplary embodiment of the present subject disclosure.

(89) FIG. 28A shows an exploded perspective view of a closed basket, according to an exemplary embodiment of the present subject disclosure.

(90) FIG. 28B shows an exploded side view of a closed basket, according to an exemplary embodiment of the present subject disclosure.

(91) FIG. 29A shows a shadow perspective view of an air escape disk, according to an exemplary embodiment of the present subject disclosure.

(92) FIG. 29B shows a top view of an air escape disk, according to an exemplary embodiment of the present subject disclosure.

(93) FIG. 30A shows a shadow perspective view of a flow control disk, according to an exemplary embodiment of the present subject disclosure.

(94) FIG. 30B shows a top view of a flow control disk, according to an exemplary embodiment of the present subject disclosure.

(95) FIG. 30C shows a side view of a flow control disk, according to an exemplary embodiment of the present subject disclosure.

(96) FIG. 31A shows a perspective view of an end cap having air windows on the side, according to an exemplary embodiment of the present subject disclosure.

(97) FIG. 31B shows a perspective view of an end cap having air windows on the top, according to an exemplary embodiment of the present subject disclosure.

(98) FIG. 32A shows a perspective view of a tapered container, according to an exemplary embodiment of the present subject disclosure.

(99) FIG. 32B shows a side view of a tapered container, according to an exemplary embodiment of the present subject disclosure.

(100) FIG. 32C shows a top view of a tapered container, according to an exemplary embodiment of the present subject disclosure.

(101) FIG. 33 shows an exploded side view of a basket, according to an exemplary embodiment of the present subject disclosure.

(102) FIG. 34A shows a perspective view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(103) FIG. 34B shows a side view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(104) FIG. 34C shows a top view of a basket assembly, according to an exemplary embodiment of the present subject disclosure.

(105) FIG. 35A shows a perspective view of a countertop brew system, according to an exemplary embodiment of the present subject disclosure.

(106) FIG. 35B shows a side view of a countertop brew system, according to an exemplary embodiment of the present subject disclosure.



- (107) FIG. 36A shows a perspective view of a countertop brew system, according to an exemplary embodiment of the present subject disclosure.
- (108) FIG. 36B shows a side view of a countertop brew system, according to an exemplary embodiment of the present subject disclosure.
- (109) FIG. 36C shows a perspective view of a conical cap, according to an exemplary embodiment of the present subject disclosure.
- (110) FIG. 37 shows a perspective view of a countertop brew system with removable pitcher, according to an exemplary embodiment of the present subject disclosure.
- (111) FIG. 38 shows a perspective view of a countertop brew system with automated control, according to an exemplary embodiment of the present subject disclosure.
- (112) FIG. 39A shows testing of a closed brew system, according to an exemplary embodiment of the present subject disclosure.
- (113) FIG. 39B shows testing of a closed brew system, according to an exemplary embodiment of the present subject disclosure.
- (114) FIG. 39C shows comparison testing, according to an exemplary embodiment of the present subject disclosure.
- (115) FIG. 40A shows simulation testing a closed system at 30 seconds, according to an exemplary embodiment of the present subject disclosure.
- (116) FIG. 40B shows simulation testing an open system at 30 seconds, according to an exemplary embodiment of the present subject disclosure.
- (117) FIG. 41A shows simulation testing a closed system at 1 minute, according to an exemplary embodiment of the present subject disclosure.
- (118) FIG. 41B shows simulation testing an open system at 1 minute, according to an exemplary embodiment of the present subject disclosure.
- (119) FIG. 42A shows simulation testing a closed system at 5 minutes, according to an exemplary embodiment of the present subject disclosure.
- (120) FIG. 42B shows simulation testing an open system at 5 minutes, according to an exemplary embodiment of the present subject disclosure.
- (121) FIG. 43A shows simulation testing a closed system at 10 minutes, according to an exemplary embodiment of the present subject disclosure.
- (122) FIG. 43B shows simulation testing an open system at 10 minutes, according to an exemplary embodiment of the present subject disclosure.
- (123) FIG. 44A shows simulation testing a closed system at 20 minutes, according to an exemplary embodiment of the present subject disclosure.
- (124) FIG. 44B shows simulation testing an open system at 20 minutes, according to an exemplary embodiment of the present subject disclosure.
- (125) FIG. 45A shows simulation testing a closed system at 30 minutes, according to an exemplary embodiment of the present subject disclosure.
- (126) FIG. 45B shows simulation testing an open system at 30 minutes, according to an exemplary embodiment of the present subject disclosure.

#### DETAILED DESCRIPTION OF THE SUBJECT DISCLOSURE

(127) The present subject disclosure addresses the shortcomings of conventional cold coffee or tea brewing systems. The present disclosure provides the user options to brew with either single serve coffee cups or free coffee grounds. Existing cold brew baskets require the user to shake, stir or pour the water over the grounds to moisten. The present subject disclosure features air escape filters which eliminate the need for any additional agitation or stimulation. Numerous coffee and tea basket designs exist in the cold brew and hot brew market, such as open immersion baskets, pour over, drip, and pressurized water such as espresso. The present closed basket combines the benefits of an immersion basket with that of the pour over or drip devices. The present systems also produce a cold-brew coffee in which the high quality (defined as high dissolved solids) coffee is efficiently

separable from the brewing water, and later drawn off without being diluted after mixing.

(128) Devices, systems, and methods according to the present subject disclosure allow users to create cold brew coffee for a single serving using the popular disposable single serve coffee cups. The mug portion of the subject disclosure may be used for other general drinking purposes, including traditional hot coffee, although its unique design and intended purpose is to house the novel brewing basket assembly, which makes use of disposable single serve coffee cups. The mug portion has been designed to be compatible with existing single-serve coffee brewing machine cups, such as, for example, the K-CUP by KEURIG, and others.

(129) The present subject disclosure allows users many options and capabilities to brew coffee or tea for personal use. The present subject disclosure provides the ability for users to make cold brew coffee using disposable single serve coffee cups. Further, users can make personal cold brew beverages using coffee or tea grounds designed for hot brew, gravity fed, traditional coffee machines. Users are also able to mix flavors available in disposable single serve coffee cups. Users do not have to shake, stir, or pour water over coffee grounds using the present system, thereby simplifying the process. Users have the ability to naturally brew the highest quality beverage possible (as defined by high percentage of dissolved solids) through natural density-driven gravitational currents of the water, particularly in cold water, without the use of external pressurized water by maximizing the benefits of Le Chatelier's Principle. Other unique attributes of the present subject disclosure will be appreciated by one having ordinary skill in the art and are within the scope of the present disclosure.

(130) The present subject disclosure shows by examples uniquely designed coffee and tea baskets and their ability to brew coffee or tea from any grounds in a portable mug device. The present subject disclosure also describes unique coffee and tea filters and the ability to filter out air and other gases trapped in the filter contents. The devices, systems and methods according to the present subject disclosure describe coffee and tea brewing methods and the ability to maximize high quality (high dissolved solids) naturally. However, the present techniques are not limited to coffee or tea but may be used with any other material that produces a beverage. Other examples include, but are not limited to, natural or artificial flavored drinks, protein drinks, medicines, etc. The use of coffee and tea are mere examples, and the full scope of the use of the present techniques would be appreciated by one having ordinary skill in the art after consideration of the present disclosure.

(131) Devices, systems and methods according to the present subject disclosure provide novel techniques to brew beverages (e.g., coffee, tea, etc.) at home, office, travel, commercial settings, etc. The versatility of the present systems and methods are presented by the examples shown and described. It should be noted that each example has been shown with unique or distinct features or elements. However, such features or elements are not limited to the specific example shown, but may be interchanged or substituted into other examples shown and described. There are countless variations and combinations of the elements as described in the examples below, and for sake of brevity, only a few examples are presented. One having ordinary skill in the art would appreciate that elements of one example may be interchanged or substituted into other examples to produce a unique desired combination, which are within the scope of the present disclosure.

(132) The examples presented herein provide for four general classes of the present subject disclosure: (1) a reusable beverage mug (FIGS. 1-11); with various additional exemplary embodiments (FIGS. 12-23); (2) an open basket mug (FIGS. 24-26); (3) a closed basket mug (FIGS. 27-31); with various additional exemplary embodiments (FIGS. 32-34); and (4) countertop models for personal/commercial environments (FIGS. 35-38). Finally, FIG. 39 shows testing results, and FIGS. 40-45 show various fluid simulations.

(133) The first class of brew devices, systems, and methods (FIGS. 1-11) includes a reusable coffee mug for making cold-brew beverages using single serve coffee/tea cups. An exemplary device uses one or two single serve coffee cups, pierced through two holes in the bottom and four slices on top,

housed in a filtered assembly attached to the mug's lid, then submerged in water for a period of time. The user can use the mug with or without the assembly remaining inside after brewing. The assembly can be removed and the mug used generally for any standard beverage.

(134) The second class of brew devices, systems, and methods (FIGS. **12-14**) includes an open-basket design which allows the user to cold-brew coffee using traditional coffee grounds in a portable mug. An additional air escape filter allows users to submerge the open basket into water without the need to shake, stir, or directly pour water over the grounds to moisten them.

(135) The third class of brew devices, systems, and methods (FIGS. **15-19**) is a closed basket design which allows users to utilize and maximize Le Chatelier's Principle, to naturally brew the highest total dissolved solids (TDS) possible without the use of external pressurized water, and without immersion in the reaction products, which function to slow the continued reaction down. All chemical reactions and physical changes are reversible by nature. Statistically, they tend to shift more one way or another based on if they are a spontaneously endothermic or exothermic. The dissolving of coffee reactants is a spontaneously endothermic process that seeks to remove heat from the surroundings. The equation can be simplified to:

Heat+Water+Coffee Solublescustom characterTDS (Total Dissolved Solids)+Water.

(136) TDS, which includes the caffeine compound, can be maximized by shifting the equilibrium further to the right into making more products. The present closed immersion design maximizes the three equilibrium factors of concentration, pressure, and temperature. By maximizing these equilibrium factors, according to Le Chatelier's Principle, the reaction will shift to making more TDS. Experimental data has shown that the present design does slow the reaction down for temperatures that approach freezing, with maximum TDS occurring at temperatures slightly above freezing. It has been shown that the addition of heat creates convection currents that disrupt the deposition of TDS on the bottom of the container, despite the increased reaction rate benefitting the rate at which TDS are generated.

(137) The present system also enhances the quality of the brewed beverage by utilizing Toricelli's Law to enhance flow during brewing. Toricelli's Law is a fluid dynamics concept that relates the amount of time it takes to drain a "tank" (simulated by the internal basket containing high-density coffee products) to the vertical displacement of the center of gravity of the fluid inside, given that the tank opening is at the bottom. According to conservation of energy for a free-falling object:

$$(138) mgh = \frac{1}{2}mv^2$$

The exit velocity of the fluid can thus be proportional to the square root of the vertical displacement, h, according to:

$$v=\sqrt{\text{square root over } (2gh)}$$

The present subject disclosure in certain embodiments uses a basket that has a conical shape, which minimizes its volume near the exit. It also constricts flow at its exit causing accumulation of coffee products. These two features work together to force accumulated coffee products to be displaced upward, maximizing h and thus v.

(139) A fourth class of brew systems and devices is a higher volume countertop design which may be used for residential/commercial environments and scales up the gravity-driven brewing mechanism found in the single serve coffee cup mug design and allows high-TDS coffee to be drawn off the bottom of a large holding container, with minimal disruption to and dilution of the high TDS coffee present at the bottom of the holding container.

(140) FIGS. **1-11** describe a reusable portable beverage mug, which is especially designed for existing disposable, single use pre-packaged cups containing tea or coffee.

(141) FIGS. **1A-1E** show various views of a beverage container **100** according to the present subject disclosure. The container **100** includes a mug portion **101** and a lid **102**. FIG. **1A** shows a closed device with a container **101** and lid **102**. FIGS. **1B** and **1C** show a device with just the container **101** and no lid. FIG. **1D** shows a top view of the container **101** with no lid. FIG. **1E** shows a top view of the container **101** with a lid **102**.

(142) An internal basket assembly **130** is disposed within the container **101** and includes modular components, as will be described in more detail below. The modular components include a spacer **140**, an upper slicer **150**, upper holder **160**, lower slicer **170**, and lower holder **180**. Upper holder **160** contains an upper cup **151**, and lower holder **180** contains a lower cup **171**.

(143) The “max fill line” **111**, which may be, for example, a 19.3 fluid ounce maximum fill level, accounts for the user filling the mug to the ideal twelve fluid ounce line **113**, added to the 7.3 fluid ounces of displacement due to the internal assembly and two air-filled single serve coffee cups **151**, **171**. The “minimum fill line” **112**, which may be, for example, a 14.24 fluid ounce minimum fill level, accounts for the user filling the mug to the “strong coffee” ten fluid ounce line **114**, added to the 4.24 fluid ounces of displacement due to the internal assembly and two water-filled single serve coffee cups **151**, **171**, and is positioned such that water will still sit above the upper holder filtration windows **144** during brewing. In use, the internal basket **130** is removed and the container **101** is filled with a fluid (e.g., water) to a point including and between low mark **114** and high mark **113**. Then the internal basket assembly **130** is positioned within the container **101**, thereby displacing the fluid that was poured into the container **101**. The low fill mark **114** to high fill mark **113** range results in a container **101** fill of low mark **112** to high mark **111**, respectively.

(144) FIGS. 2A-2C show the container/mug portion **101** from a side perspective (FIG. 2A), a side (FIG. 2B), and a top view (FIG. 2C). One or markers **113**, **114** may indicate the fill level of the mug before insertion of the internal basket **130**. The mug threads **119** engage the mug lid threads **109** (shown in FIG. 3A).

(145) FIGS. 3A, 3B, 3C, and 3D show a mug lid **102** in a shadow perspective view, shadow top view, bottom view, and top view, respectively. The lid **102** includes a thick, tall lip **103** for a “substantial feel” and to allow clearance of the user's nose while sipping.

(146) The lid's vertical channels **104** engage and align the spacer tabs **142** during assembly (See FIG. 7) in order to connect the basket assembly **130** to the lid **102**. When engaged and twisted clockwise, the spacer tabs **142** slide through the lid horizontal channels **105** until they hit a stop. The lid opening **106** allows both sipping and use of a straw. It cuts slightly into the bottom edge of the lip **103** to create a low point for coffee to flow to during use. It is sufficiently tall to allow a straw to sit at an angle and reach the bottom of the mug **101** when inserted and has sloped walls to further allow for a straw's deflected angle of insertion.

(147) There is clearance **107** between the lid opening **106** and the slotted portion of the lid below to allow for movement of or stirring with a straw when used (and assemblies removed). See FIGS. 1C, 3C, and 3D. The lid vent **108** allows air to enter the container **101** as coffee exits. The lid vent **108** is helpful if the user's lips make an air-tight seal against the contours of the mug lid **102**. The lid vent **108** also functions to equalize internal/external air pressure.

(148) The mug threads **119** on the container **101** engage the mug lid threads **109** on the lid **102**. The mug lid **102** threads onto the mug **101** to prevent misalignment or banging of the internal assembly **130** when it is attached to the mug lid **102** while the lid **102** is either being screwed into or unscrewed from the mug **101**.

(149) FIGS. 4A-4F show the upper slide portion **121** and lower slide portion **126** of the closing mechanism within the lid **102**. More specifically, FIGS. 4A-4C show a top side perspective, bottom side perspective, and bottom view, respectively, of the upper slide portion **121**. Further, FIGS. 4D-4F show a top side perspective, top view, and side view, respectively, of the lower slide portion **126**. The upper slide portion **121** is to be operated by a user's finger as the user opens/closes the mug opening **106**. The upper side portion **121** has a slide cut-out **122** which allows for insertion of the lower closing mechanism tab **127**. Keyway **123** engage and snap onto the nub **128** on the lower closing mechanism tab **126**. The lower slide portion **126** slides over the mug opening **106** when operated by the user via the upper slide portion **121**. Further, tab **129** is used to assist in holding lower closing mechanism tab **126** during assembly. Other sliding mechanisms are possible and within the scope of the present disclosure.

(150) FIGS. 5A, 5B, and 5C, show a side perspective, side, and top view, respectively, of the basket assembly **130**. The basket assembly comprises of several modular and separable components, including a top spacer **140**, an upper/top holder **160**, and a lower/bottom holder **180**. Each of the spacers **140**, and holders **160**, **180** have a plurality of windows, including larger and smaller windows. The smaller windows (e.g., **152**, **172**) are covered with filters (not shown) to allow water to penetrate the interior of the basket **130** and interact with the coffee grinds or tea leaves housed therein. The larger windows (e.g., **164**, **166**) act as cut outs to reduce the volume of material within the device.

(151) FIGS. 6A, 6B, and 6C show an exploded bottom perspective, an exploded side, and a top view, respectively, of the basket assembly **130**. The spacer **140** has a top slicer **150** connected to its underside. The top slicer **150** engages with a top portion of a first disposable coffee cup **151**, which is housed within the top holder **160**. A bottom slicer **170** is connected to the underside of the top holder **160**, which engages with a top portion of a disposable coffee cup **171**, which is housed within the bottom holder **180**. This set up allows a user to have one or two coffee cups **151**, **171**, having the same or different flavors. Each component is modular and may be replaced or exchanged as needed, and will be described in detail individually.

(152) FIGS. 7A, 7B, and 7C show a side perspective, side and top view of the spacer **140** and upper slicer **150**. The upper slicer **150** is shown attached to the internal spacer **140**. A function of the spacer **140** is to displace the internal assembly **130** vertically downward to keep the water level above the top level of the upper coffee cup **151**, and allow water level fluctuation during initial submersion, when the single serve coffee cups **151**, **171** are filled with air. The spacer **140** allows water to reach a maximum level **111** (due to air-filled single serve coffee cups **151**, **171**) as well as a minimum level **112** (due to water-filled single serve coffee cups **151**, **171**) that is above the upper holder windows **164** during brewing, subject to the user properly filling the mug to the twelve fluid ounce fill line **114**.

(153) During brewing, water will move through the spacer windows **144** and the slicer windows **152**. The spacer tabs **142** engage the mug lid's vertical channels **104** of the mug lid **102** during assembly, and act as a locking mechanism after they are rotated clockwise through the lid's horizontal channels **105** and hit a stop. The spacer **140** has a round top portion **141** and a round bottom portion **143**, which are connected through a tapered intermediate portion **145**. The tapered portion **145** and the smaller round top **141** allow for coffee to flow through the lid opening **106** if the user chooses to leave the internal assembly **130** inside the mug **101** while drinking.

(154) The single serve coffee cup foil slice windows **152** sit horizontally over the slicers **153** in the foil after assembly and allow water to flow downward into the single serve coffee cups **151**. The spacer collar **143** engages the circular alignment feature **168** in the upper holder **160** and allows for a more rigid feel during assembly, especially if assembled without a single serve coffee cup **151**, which creates compressed rigidity when present. The filtration mesh collar (not shown in this figure, but which accommodates screen **334** in FIG. 16) allows for the manual installation of a circular disk **147** of stainless-steel mesh, positioned horizontally over the foil slice window (not shown but similar to **174** in FIG. 9C) and slicers **153** in the single serve coffee cup **151** after assembly. The circular disk mesh **147** prevents buoyant, loose coffee grounds from escaping during brewing. The spacer air slots **146** allow for trapped air in the upper spacer to escape and exit through the lid air slots **106A**, and eventually the lid vent **108** and the lid opening **106**. Along with the slicers **153** is an optional center punch mechanism **157**. The center punch **157** works along with the knives/slicers **153** to get the grounds fully soaked and allow trapped air to escape.

(155) FIGS. 8A, 8B, and 8C show a side perspective view, side view, and upper view, respectively, of the upper holder **160** with lower slicer **170** attached thereto. The upper holder **160** has many of the same structural features as the lower holder **180**. The vertical channels **161** of the upper holder **160** provide stability as the user pushes (using the spacer **140**) the single serve coffee cups **151** down onto the stainless steel hollow needles **165** embedded in the holder exit holes. The vertical

channels **161** engage the slicer tabs **156** during assembly. When the slicer tabs **156** reach the bottom of the vertical channels **161**, they are rotated clockwise through horizontal channels **163** until they hit a stop. The holder **160** outer windows **162** allow water to enter.

(156) After assembly, the holder outer windows **162** align with the slicer inner windows **152** and have the same vertical and angular dimensions. Cut-outs **164** in the upper holder minimize the overall volume of the assembly and the material requirements during manufacturing. The single serve coffee cup **151** access window **166** allows the user to grasp the single serve coffee cup **151** with fingers on both sides in order to remove it after brewing is complete. The shelf **177** provides support to the single serve coffee cup **151** as it is compressed and sliced open by the slicer knives **153**. The clearance cutout **167** allows space for a protruding collar that often occurs below the rim of the single serve coffee cup **151**, making for easier assembly.

(157) FIGS. **9A**, **9B**, and **9C** show a side perspective, side view, and top view, respectively, of the lower slicer **170**. The slicer filtration windows **172** align (after assembly) with the holder filtration windows **182** and have the same vertical and angular dimensions. The slicer tabs **176** function in alignment when engaged with the vertical channels **181** and as an assembly locking mechanism in the horizontal channels **183** after assembly. The single serve coffee cup foil slice windows **174** sit horizontally over the slices made in the foil by **173** after assembly and allow water to flow into the single serve coffee cups **171**. In some embodiments, a slicer removal hole **175** is used instead of the center punch **177**, with the hole **175** having a larger radius than the holder removal hole **186** and allows the user to poke a finger to a greater depth than the holder removal hole **186** when removing the single serve coffee cups **171** after brewing. The slicer knives **173** are positioned radially to cut through the center of the single serve coffee cup foil slice windows **174** and are manually plunged through the single serve coffee cup foil during assembly. They create angular cuts when the user twists the slicer during assembly, as will be described in detail in FIG. **11**. Along with the slicer knives **173** is an optional center punch mechanism **177**. The center punch **177** works along with the knives **173** to get the grounds fully soaked and allow trapped air to escape.

(158) FIGS. **10A**, **10B**, and **10C** show a side perspective view, side view, and upper view, respectively, of the lower holder **180**. The vertical channels **181** of the upper and lower holders provide stability as the user pushes the single serve coffee cup **171** down onto the stainless steel hollow needles embedded in the holder exit holes **185**. The vertical channels **181** engage the slicer tabs **176** of the lower slicer **170** during assembly. When the slicer tabs **176** reach the bottom of the vertical channels **181**, they are rotated clockwise through horizontal channels **183** until they hit a stop. The holder outer windows **182** allow water to enter/exit. After assembly, the holder outer windows **182** align with the slicer inner windows **172** and have the same vertical and angular dimensions. Cut-outs **184** in the holder **180** minimize the overall volume of the assembly and the material requirements during manufacturing. The single serve coffee cup **171** access window **189** allows the user to grasp the single serve coffee cup **171** with fingers on both sides in order to remove it after brewing is complete. The shelf **187A** provides support to the single serve coffee cup **171** as it is compressed and sliced open by the slicer knives **173**. The clearance cutout **187B** allows space for a protruding collar that often occurs below the rim of the single serve coffee cup **171**, making for easier assembly.

(159) The slicer windows **172** align (after assembly) with the lower holder outer windows **182** and have the same vertical and angular dimensions. The slicer tabs **176** function as alignment features when engaged with both the circular alignment feature **188** and the vertical channels **181** of the lower holder. They act as locking mechanisms in the lower holder's **180** horizontal channels **183** after assembly. The slicer knives **173** are positioned radially to cut through the center of the single serve coffee cup and are manually plunged through the single serve coffee cup foil during assembly. The slicer knives **173** create angular cuts when the user twists the slicer **170** clockwise during assembly. The lower holder has angular cut-outs **184** to minimize displacement volume during brewing and material during manufacturing.

(160) FIGS. **11A**, **11B**, and **11C** show side perspective view, side view, and top view, respectively, of slicer knives **173**. The knives **173** are shown to have an inner side **173A** and outer side **173B** wherein the inner side **173A** has a smaller radius than the outer side **173B** to minimize friction and binding forces during rotational cutting.

(161) FIGS. **12-23** disclose additional exemplary embodiments which have different features and configurations as those shown and described in FIGS. **1-11**. For sake of brevity, each of the embodiments of FIGS. **12-23** will not be described in detail, but only the features which are different will be shown and described in the figures and associated discussion.

(162) FIGS. **12A**, **12B**, and **12C** show side perspective view, exploded side view, and top view of a closed basket **130**. This embodiment allows for independent twisting of the slicers **150**, **170** similar to a gas can collar instead of having the slicers fixed to the upper holders **140**, **160**, respectively.

(163) FIGS. **13A**, **13B**, and **13C** show side perspective view, exploded side view, and top view of a closed basket **130**. This embodiment presents a simple push down and latch design with hollow needles to allow coffee flow, with no twisting necessary.

(164) FIGS. **14A**, **14B**, and **14C** show side perspective view, exploded side view, and top view of a closed basket **130**. This embodiment is a simplified design with no top windows in the upper slicer **150**, but still allows water to flow down through the upper spacer assembly **140**.

(165) FIGS. **15A**, **15B**, and **15C** show side perspective view, exploded side view, and top view of a closed basket **130**. This embodiment presents an open center design to allow the straight downward flow of brewing coffee.

(166) FIGS. **16A**, **16B**, and **16C** show side perspective view, exploded side view, and top view of a closed basket **130**. This embodiment includes knives which are recessed instead of protruding so they don't stick out of the slicers **150**, **170**.

(167) FIGS. **17A**, **17B**, and **17C** show side perspective view, exploded side view, and top view of a closed basket **130**. This embodiment includes a feature where the slicers **150** are screwed on instead of pushed down and twisted.

(168) FIGS. **18A**, **18B**, and **18C** show side perspective view, exploded side view, and top view of a closed basket **130**. This embodiment includes a feature where the slicers **150**, **170** are pushed down until a spring button engages in the holders **160**, **180**. The spring button allows release as well.

(169) FIGS. **19A**, **19B**, and **19C** show top perspective view, side perspective view, and bottom view of a lid **101**. This lid **101** with threads **101A** allows the upper holder to be threaded onto the lid **101** instead of pushed up and twisted.

(170) FIGS. **20A**, **20B**, and **20C** show side perspective view, side view, and top view of an upper holder **160**. This embodiment is similar to prior described designs, except that the tabs ride along the ramp **170A** in the slicers **170** until they reach the slot and engage.

(171) FIGS. **21A**, **21B**, and **21C** show side perspective view, side view, and top view of an upper holder **160**. This embodiment is similar to prior described designs, except that it only has two tabs **176** on the slicer **170** that ride along threads in the holders.

(172) FIGS. **22A**, **22B**, and **22C** show side perspective view, side view, and top view of an upper holder **160**. This embodiment is similar to prior described designs, except that the positions are reversed, with the tabs on the holders, and the slots **170B** on the slicers **170**.

(173) FIGS. **23A**, **23B**, and **23C** show side perspective view, side view, and top view of top spacer **140**. This embodiment is similar to prior described designs, except that to go along with the threaded lid, the spacer **140** is threaded and screws on to the threaded lid.

(174) FIGS. **24-26** describe a brew device having an open basket assembly used to brew conventional coffees and teas in a holding container.

(175) FIGS. **24A**, **24B**, and **24C** show side perspective, side, and top view of a brewing device **200** having an open filtered basket assembly **230**. The device has a container **201** and a lid **202**, similar to that described in FIG. **1**. Further, and similarly to the device shown and described in FIG. **1**, there exists a range of brewing volumes, from a minimum fill level of **214** to a maximum fill level

of **213**. These can correspond to, for example, 12 ounces to 16 ounces. The minimum brewing depth **214** indicates the water level after submersion for the twelve fluid ounce minimum brewing option where the basket windows **232** remain necessarily below the water level.

(176) FIGS. **25A**, **25B**, and **25C** show side perspective, side, and top view of an open filtered internal basket **230**. It includes a grip portion **231** that both allows the user to handle and twist the basket as it is inserted into the mug lid **202**. It displaces the basket downward so that its upper windows **232** are submerged when the user has chosen the twelve fluid ounce brewing option and the water level sits at the basket minimum brewing depth **233**. An internal stainless-steel filtration screen (not shown) holds buoyant coffee grounds under water during brewing. The basket has large, open windows **234** that maximize water's access to the coffee grounds inside during brewing. An air escape disk **236** sits directly above the deposit of coffee grounds, allowing air to escape. Any coffee grounds that escape through the disk **236** are trapped behind the windows **232** and the screened surface directly above. The basket threads **239** engage the basket lid threads **246** during assembly. When the basket **230** is inverted, the twelve and sixteen fluid ounce coffee ground fill lines **237** and **238**, respectively, indicate to the user where to fill with coffee grounds for those respective brewing options.

(177) FIG. **25D** shows a side perspective view of an exemplary coffee grinds or tea leaf holder **250** which is positioned within the basket **230**. The holder **250** has a round top side **251**, a round bottom side **252**, and a cylindrical body **253**. The top side **251** and cylindrical body **253** have a filter mesh (not shown). The bottom side **252** is open. In use, the holder **250** is filled with coffee grinds or tea leaves as desired, then inserted into internal basket **230**. A basket lid **240** (see FIG. **14**) is then threaded onto the basket **230**, thereby securing the coffee grinds or tea leaves within the holder **250**. One or more slits and/or holes **254** may be made at the top of the holder **250** to facilitate the escape of trapped air in the brewing steps.

(178) FIGS. **26A**, **26B**, and **26C** show side perspective, side, and top view of an open filtered internal basket lid. It has a cross-bar **241** for tightening and loosening and includes two large openings **242** that allow for maximum water flow during brewing. The basket lid threads **246** engage the basket threads **239** during assembly. A temporary filtration screen shelf **245** is included to allow manual installation of a disk of stainless-steel filtration mesh.

(179) FIGS. **27-31** describe a brew device having a closed basket assembly which uses Le Chatelier's Principle to produce a high quality beverage.

(180) FIGS. **27A**, **27B**, and **27C** show side perspective view, exploded side view, and top view of a closed internal basket **330**. Such an embodiment may be used for personal or commercial purposes. The basket assembly **330** includes a top portion **331** and a bottom portion **332**, which are threaded together. The closed internal basket **330** does not have the open windows **231**, **234** of the open basket design shown in the prior examples. Instead, there is a limited and controlled port area where water can access the coffee positioned within the basket **330**, and a limited port where brewed coffee may exit the basket **330**. This port area is similar in design to that described in the examples shown in FIGS. **8** and **10**.

(181) FIGS. **28A** and **28B** show an exploded side perspective view of a mug assembly **300**, and an exploded side view of a mug assembly **300**, respectively. The closed basket assembly **330** is shown. A top spacer **332** has a tab portion **331** that allows locking into a lid (not shown, but similar to those shown above in prior figures). The spacer **332** has a thread portion **333** that mates with a corresponding thread portion on the basket body portion **340**. A horizontal stainless-steel filtration screen **334** sits vertically above the filtration windows, which contain stainless-steel filtration screens **335**. An air escape disk **336** (see FIG. **29**) fits into the bottom of the upper part **332** of the assembly and allows air to escape from the coffee grounds while minimizing the passage of cold brew-sized grounds. The conical filtration cartridge **338** is made from stainless-steel filtration screen and is supported by channeled ring end pieces **337**, **339**, with the bottom **339**, smaller ring spanned by a stainless-steel filtration screen.



(182) The closed basket housing **340** has a conical shape and recessed internal vertical ribs **340A** which allow dissolved coffee products to fall vertically downward along the channels formed by the ribs as they are created, making room for new coffee products to be created. It has four exit holes **342** in the flow control disk **341**, whose total cross-sectional area can be optimized in order to minimize friction, and/or maximize flow according to Toricelli's Law. The exit holes **342** are covered by small stainless-steel filtration screen disks **343** which sit in recessed grooves around the perimeter of the holes **342** on the bottom. A flow control disk **341** sits below the exit holes **342** and can be rotated to increase or decrease exposure to the basket's exit holes, allowing the user to regulate the rate at which coffee products exit the basket, and thus control the rate at which coffee is brewed.

(183) FIGS. **29A** and **29B** show a shadow side perspective view and a top view of an air escape disk **336**, shown in place in FIG. **28**. The air escape disk **236** (see FIG. **25A**) is similar to the air escape disk **336** and shown in detail in FIG. **29**. The disk **336** has arc-shaped apertures **336A** and a center hole **336B** which allow air to escape but which also minimize the passage of cold brew-sized grounds. The dimensions of the cuts **336A** and center hole **336B** have been determined experimentally. For example, the apertures **336A** may be 1.59 mm but could range size from 1 mm to 2 mm. The center hole **336B** may be, for example, 6.46 mm, but could range from 3.18 mm to 9.53 mm.

(184) FIGS. **30A**, **30B**, and **30C** show side perspective view, top view, and side view of a flow control disk **341**. It has a protruding radial finger tab **344** that allows the user to manually rotate it. It has four exit holes **342** in the bottom, whose total cross-sectional area can be optimized in order to minimize friction and/or maximize flow rate and which match in size and position the exit holes in the bottom of the closed basket housing **340**. The disk **341** is mounted to the housing **340** by a protruding axle nub **345** that fits into a recessed hole of the same shape in the bottom center of the closed basket housing **340**.

(185) Applicants have shown through numerous experiments that a closed basket system with a very limited brewed exit port openings (e.g., two exit holes **342** open) results in a high velocity brewing of coffee, and production of a high quality (high TDS) beverage, as measured by samples at the bottom of the container. It appears that a larger size for the exit hole **342** will decrease the speed and cause TDS to decrease, while a smaller size for the exit hole **342** will increase the speed and cause TDS to increase. The hole sizes may be, for example, 6 mm, but could range in size using the rotatable controller **341** anywhere from 1 mm to 9.53 mm.

(186) FIGS. **31A** and **31B** provide side perspective views of two further exemplary embodiments of alternative lids **350** for housing basket **340**. Each lid **350** has one or more holes **351**, with filters positioned therein (not shown). A thread **352** allows the lid **350** to be threaded into the bottom portion of basket housing **340**. Fluid windows **353** may be positioned horizontally (FIG. **31A**) or vertically (FIG. **31B**) to facilitate movement of air and fluids, and to drain the internal vertical ribs similar to **340A** (see FIG. **28**) which are the interior surface of basket housing **340** (see FIG. **33**) to thereby enhance the rate of brewing.

(187) FIGS. **32-34** disclose additional exemplary embodiments which have different features and configurations as those shown and described in FIGS. **24-31**. For sake of brevity, each of the embodiments of FIGS. **32-34** will not be described in detail, but only the features which are different will be shown and described in the figures and associated discussion.

(188) FIGS. **32A**, **32B**, and **32C** show side perspective view, side view, and top view of a closed internal basket **330**. This embodiment is wider at the bottom to allow for easier filling of coffee/tea.

(189) FIG. **33** shows a side view of a closed internal basket **230**. This embodiment includes an air escape disk **336** and O-ring **337** that cap the ends of a cylinder of screen that can be filled from the bottom and capped with lid **350**. O-ring **337** aligns vertically over fluid windows **353** (see FIG. **31**) to allow fluid through fluid windows **353**. The exemplary lids **350** shown in FIG. **31** are used as optional lids for the basket **330** shown in FIG. **33**.

(190) FIGS. **34A**, **34B**, and **34C** show side perspective view, side view, and top view of a closed internal basket **330**. This embodiment has an inner filter **359** that runs down through the center of the basket **340** to allow water flow through the center in addition to the sides.

(191) FIGS. **35-38** describe various exemplary countertop versions of the brew systems, to be used for higher volume personal/commercial environments.

(192) FIGS. **35A** and **35B** show a side perspective view, and a side view, respectively, of a high volume beverage dispensing system **400**. The closed basket assembly **430** is shown mounted inside a high volume brewing container **401**. A drain valve **402** is mounted on the side, above which is mounted a baffle plate **403** meant to disrupt the formation of a vortex (and thus unwanted mixing) when the brewed beverage in the container **401** is being dispensed, either fully or partially, as servings of the brewed beverage are drawn off. Further, a sloped decline **404** allows a smooth flow of brewed coffee from the internal basket assembly **430** to the bottom of the brewing container **401** while minimizing the turbulence of the layer of dense brewed coffee resting at the bottom of container **401**. A handle **430** provides for an easy removal of the basket assembly **430** for re-filling with coffee, washing, repairing, etc. The rate of flow of the brewed beverage is controlled by flow control disks **450** acting as end caps, similar to those described in FIGS. **18-19**.

(193) FIGS. **36A** and **36B** show a side perspective view, and a side view, respectively, of a high volume beverage dispensing system **500**. The closed basket assembly **530** is shown mounted inside a high volume brewing container **501**. A drain valve **502** is mounted on the side, and the shape of the bottom **503** of the container is for draining purposes. A vortex breaker **535** is located at the tip of the cone to disrupt the formation of a vortex (and thus unwanted mixing) when the brewed beverage in the container **401** is being drained, either fully or partially, as servings of the beverage are drawn off. Further, the conical plate **503** has a sloped decline **504** which allows a smooth flow of brewed coffee from the internal basket assembly **530** to the bottom of the brewing container **501** through tubing **505** while minimizing the turbulence of the layer of dense brewed coffee resting at the bottom of container **501**. A handle **531** provides for an easy removal of the basket assembly **530** for re-filling with coffee, washing, repairing, etc. A vertical siphon tube **505** is attached to the basket's exit hole, and functions to both directly deposit coffee products on the bottom of the container, as well as maximize the pressure differential between the exit and the intake windows at the top, thus maximizing flow. A horizontal, circular impingement plate **536** at the bottom of the siphon dissipates kinetic energy of exiting coffee products (impingement jet) and thus minimizes turbulence, and also functions to restrict turbulence during draining through the drain valve **502**. A vortex breaker **535** is included immediately at the tip of the conical plate to disrupt the formation of vortices during draining.

(194) FIG. **36C** provides a perspective view of an optional conical cap **532** which may be screwed onto the basket **530** through thread **534**. This conical cap uses Toricelli's Law to minimize its volume near the exit. It also constricts flow at its exit causing accumulation of coffee products and maximizing exit velocity according to Toricelli's Law. These two features work together to force accumulated coffee products to be displaced upward, thereby maximizing the downward flow of brewed coffee, and increasing TDS. The apertures **533** about the periphery of the end cap allow for the continuous flow of downward drifting brewed beverage as it flows down the internal ribs **530A** of the basket **530**. Although shown specifically in this exemplary embodiment, the conical end cap **532** may be used in any of the embodiments shown and described here.

(195) FIG. **37** shows another exemplary embodiment of a high volume beverage brew system **600** according to the present subject disclosure. In this system **600**, there is a base **601** and a reservoir **602**. The base **601** may contain a heating element **603** to provide optional heat to the bottom portion of a removable pitcher **605** having handle **606**. The removable pitcher **605** serves as the brewing container and is adapted to house the internal basket **630**. The removable pitcher **605** may be used to easily dispense brewed beverage from within the container. Optionally, a drain **607** may be used to drain out unbrewed or underbrewed fluid from the pitcher **605** before removing the

pitcher **605** from the system **600**. This way, the remaining beverage in the pitcher **605** is fully brewed and will not be diluted from underbrewed fluid resting atop the brewed beverage. This system **600** may be of any size and may be used for residential or commercial purposes.

(196) FIG. **38** shows another exemplary embodiment of a beverage brew system **700** according to the present subject disclosure. In this system **700**, there are two compartments, an unbrewed fluid reservoir **701** and a brewed beverage container **702**. A heating element **703** may be used to heat the fluid before being used to brew. A removable internal basket **730** holds grounds or leaves for brewing. The internal basket **730** has a handle **706** to facilitate in removable and replacement. A dispenser **704** dispenses brewed beverage from the brew tank **702**. Optionally, a drain **707** may be used to drain out unbrewed or underbrewed fluid from the brewed beverage tank **702**. This way, the remaining beverage in the brewed beverage tank **702** is fully brewed and will not be diluted with underbrewed fluid resting atop the brewed beverage when the beverage is dispensed through the dispenser **704**. This system **700** may be of any size and may be used for residential or commercial purposes. Because of the potentially large holding tanks, and the resultant weight, such a system may be permanently situated in a home or commercial setting.

(197) Further, the system **700** may be automated through computer control (not shown) to ensure a steady supply of readily dispensable brewed beverage. A fluid sensor may be positioned at a desired height in the brewed beverage tank **702** to initiate the brewing process by pulling water in from the storage tank **701** whenever fluid drops below the desired level in the brewed tank **702**. This ensures a steady supply of brewed beverage, which is very helpful in a commercial setting with a constant demand for the brewed beverage. A further sensor may indicate when the coffee grinds or tea leaves in the internal basket **730** should be changed to ensure a high quality beverage product.

(198) TDS testing was performed on a simulated commercial design, similar to the embodiment shown in FIG. **35**, minus the diagonal plate **404**. As shown in the data chart and visual depiction shown in FIG. **39A**, in testing aimed at creating the highest possible Total Dissolved Solids (TDS) in a simulation of a commercial version of the present design (medium sized container to maximize coffee/water ratio), it was discovered that gentle deposition of coffee products using two exit holes produced a concentration of coffee product that exceeded 3% TDS after one hour. This number decreased with time (attributed to Brownian Motion, convection currents and unintentional agitation/turbulence), but still doubled the TDS industry standard for cold brew (around 1.5%). Even after a long period of time (23.5 hours), the TDS of the coffee sitting at the bottom of the container was still significantly higher than the industry standard. The present commercial design is able to draw this coffee off without mixing it with the brewing water.

(199) As shown in the data chart and visual depiction shown in FIG. **39B**, in testing aimed at creating the highest possible TDS in the simulation of a commercial version of the present design (large sized container to maximize clean brewing water), it was discovered that gentle deposition of coffee products using two exit holes produced a concentration of coffee product that was well above the industry standard for cold brew (around 1.5%) after 7 hours. The present commercial design is able to draw this coffee off without mixing it with brewing water.

(200) As shown in the data chart and visual depiction shown in FIG. **39C**, in testing aimed at comparing the industry standard brewing method (submerged mesh bag or screened basket) to the present gravity-driven, constant flow method in the portable mug version of the design, it was discovered that in every case, the present design exceeded the TDS levels (bottom layer and mixed) of the simulated open basket industry standard design, and for tests of varying lengths.

(201) In addition to the physical testing of the present novel closed brew method, a computer program was used to differentiate the present method with that of the typical open immersion method, which is to use a mesh basket submersed in water and not enclosed in any way. The computer program used was SOLIDWORKS Flow Simulation 2020. The main variables compared were the global goals (GG) of average velocity, bulk average velocity, maximum velocity and the

minimum velocity for a reference point. A global goal is one that measures the value across the entirety of the fluid. The results of the simulation were that in every case the closed method had higher velocity values compared to the open method. The results are summarized in the table below.

(202) TABLE-US-00001 TABLE 1 GG measured between open basket and closed basket lids tall-open tall- closed basket- basket- Goal (Averaged Value) open lid two hole lid GG Minimum Velocity [mm/s] 0.00 0.00 GG Average Velocity [mm/s] 8.95 9.36 GG Maximum Velocity [mm/s] 64.78 67.20 GG Bulk Av Velocity [mm/s] 8.85 9.33

(203) Snapshots of the simulation were recorded for open and close options at the intervals of 30 seconds (FIG. 40), 1 minute (FIG. 41), 5 minutes (FIG. 42), 10 minutes (FIG. 43), 20 minutes (FIG. 44), and 30 minutes (FIG. 45). The maximum velocity global goal was chosen to best demonstrate the differences between the two methods. It should be noted that the pictures of the fluid velocity are not representative of just the coffee but of the combined water and coffee flow.

(204) An interpretation of the physical testing and the flow simulation results is as follows. The open immersion method allows the coffee to exit the basket and into the water on all sides and the bottom. The present cold brew method uses an inner mesh basket that is enclosed by a solid container. This solid container is ribbed to allow water to still access the coffee on all sides, but the solid container focuses the coffee down through the bottom of the basket. The coffee settles to the bottom of the container and is more concentrated because it wasn't able to exit through the sides. A pressure flow exists in both methods due to the coffee exiting the basket and additional water moving in to replace it. Since the coffee settles at the bottom in the closed method design, there is less resistance of water flow moving to replace the water that went into the coffee basket and exited as coffee. The less resistance would allow an increased velocity to occur in the closed basket design. Physical testing results have aligned with the flow simulation and demonstrated that the closed immersion brew method is faster at producing coffee than an open immersion method and it's more concentrated as well.

(205) The foregoing disclosure of the exemplary embodiments of the present subject disclosure has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the subject disclosure to the precise forms disclosed. Many variations and modifications of the embodiments described herein will be apparent to one of ordinary skill in the art in light of the above disclosure. The scope of the subject disclosure is to be defined only by the claims appended hereto, and by their equivalents.

(206) Further, in describing representative embodiments of the present subject disclosure, the specification may have presented the method and/or process of the present subject disclosure as a particular sequence of steps. However, to the extent that the method or process does not rely on the particular order of steps set forth herein, the method or process should not be limited to the particular sequence of steps described. As one of ordinary skill in the art would appreciate, other sequences of steps may be possible. Therefore, the particular order of the steps set forth in the specification should not be construed as limitations on the claims. In addition, the claims directed to the method and/or process of the present subject disclosure should not be limited to the performance of their steps in the order written, and one skilled in the art can readily appreciate that the sequences may be varied and still remain within the spirit and scope of the present subject disclosure.

## Claims

1. A brewing device for brewing beverages configured for use in a container using a disposable single serving container, the disposable single serving container comprise a bottom and a foil top, the brewing device, comprising: a basket assembly that comprises: a top holder that is adapted to receive the disposable single serving container; a spacer that is a lid to said top holder and holds

said top holder at a depth inside the container when the container is filled with a liquid; a top slicer comprising at least one slicer knife mounted to the underside of said spacer such that securing said spacer to said top holder, moves said top slicer in a rotational motion through the foil top of the disposable single serving container, creating an opening in the foil top of the disposable single serving container accessible by a foil slice window and at least one slicer window of the top slicer that allows gas to escape and liquid to enter the disposable single serving container when the container is filled with liquid; and a needle positioned in said top holder wherein placing the disposable single serving container in said top holder and securing said spacer on said top holder, causes said needle to penetrate the bottom of the disposable single serving container to further allow gas to escape and liquid to enter when the container is filled with liquid; a bottom holder adapted to hold a second disposable single serving container securable underneath said top holder; wherein the top holder comprises a bottom slicer connected to the underside of the top holder and configured to engage with a top portion of a second disposable coffee cup which is housed within the bottom holder.

2. The brewing device in claim 1, wherein the bottom slicer comprising at least one slicer knife positioned wherein securing said bottom holder to said top holder moves said bottom slicer in a rotational motion through the foil top of the second disposable single serving container, creating an opening in the foil top of the second disposable single serving container accessible by a foil slice window and at least one slicer window of the bottom slicer that allows gas to escape and liquid to enter the second disposable single serving container when the container is filled with liquid.

3. The brewing device in claim 2, further comprising: a second needle positioned in said bottom holder wherein placing the second disposable single serving container in said bottom holder and securing said bottom holder on said top holder, causes said second needle to penetrate the bottom of the second disposable single serving container to further allow gas to escape and liquid to enter when the container is filled with liquid.

4. The brewing device of claim 2, wherein said bottom holder having a plurality of windows to allow liquid to enter said basket assembly.

5. The brewing device of claim 2, wherein the at least one slicer window of the bottom slicer comprises the plurality of slicer windows to allow liquid to enter said basket assembly.

6. The brewing device of claim 2, wherein the at least one slicer window of said bottom slicer has a single serve coffee cup slice window.

7. The brewing device of claim 2, wherein said bottom slicer has a center punch mechanism.

8. The brewing device of claim 2, wherein said bottom holder has access windows which allows the user to grasp the disposable single serving cup.

9. The brewing device of claim 2, wherein said foil slice window of the top slicer and the bottom slicer have a filter.

10. The brewing device of claim 2, wherein the at least one said slicer window of the top slicer and the at least one said slicer window of the bottom slicer have a filter.

11. The brewing device of claim 1, wherein said spacer having a plurality of windows to allow liquid to enter said basket assembly.

12. The brewing device of claim 1, wherein said top holder has a plurality of windows to allow liquid to penetrate the interior of said basket assembly.

13. The brewing device of claim 1, wherein the at least one slicer window of said top slicer comprises the plurality of slicer windows to allow liquid to enter said basket assembly.

14. The brewing device of claim 1, wherein said top slicer has a center punch mechanism.

15. The brewing device of claim 1, wherein said top holder has access window which allows the user to grasp the disposable single serving container.

16. The brewing device of claim 1, wherein said top slicer is a screw top that is secured to the top holder by rotating the top slicer.

17. The brewing device of claim 1, wherein said bottom slicer is a screw top that is secured to the bottom holder by rotating the bottom slicer.

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